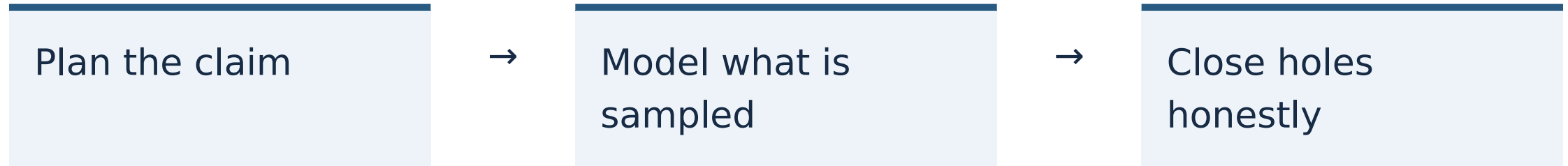


Planning and measurement

Every technique must discharge a named verification claim.



Learning outcomes

Demonstrate these decisions on a real plan row.

- Write a plan row with one closure criterion.
- Design sampling around the event of interest.
- Classify a hole with a reviewable reason.
- Explain a readiness decision using correlated evidence.

Route through the lesson

Four blocks, each ending with a decision.

- Plan the claim
- Model what is sampled
- Close holes honestly
- Read the evidence together

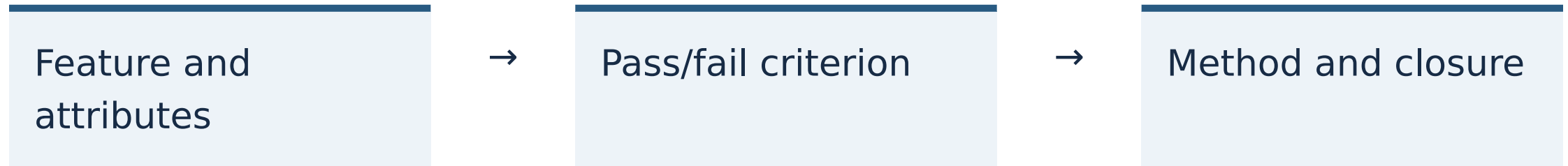
Extract behavior, not headings

Omissions in the plan become omissions in verification.

- Read interfaces, functions and architecture.
- Record forbidden and out-of-scope behavior.

A row needs one closure argument

Separate claims that require different evidence.



Keep traceability executable

Stable identifiers connect intent to results.

- Link specification, feature, tests and coverage.
- Make missing links visible in review.

Discussion: one row, two verdicts

Name the evidence that would settle the decision.

- A row claims legal bursts and correct copied data.
- Its only criterion is an assertion result.
- How should the plan change?

Coverage families have different subjects

A full structural report can miss absent functionality.



Bins belong at behavior boundaries

Convenient partitions can merge cases that behave differently.

- Choose attributes from the specification.
- Shape crosses to justified combinations.
- Sample attributes when they describe the same event.

Sampling is not checking

A filled bin needs an active correctness oracle.

- Coverage records that a modeled situation occurred.
- A checker determines whether its behavior was acceptable.

Discussion: contention at the wrong instant

Name the evidence that would settle the decision.

- Contention is sampled only when a job retires.
- The defect requires contention during streaming.
- What must the model remember?

A coverage hole is a diagnosis

Different causes require different actions.

- Check stimulus reachability and model intent.
- Distinguish unexercised behavior from impossible behavior.

An exclusion is a reviewable artifact

The report must retain the reason a hole was removed.

- Classify the exclusion.
- Record reason, owner and relevant version.
- Review it when the design or plan changes.

Risk determines depth

Good observability makes checking affordable; it does not remove risk.

- Consider consequence, novelty and specification quality.
- Agree priority and cut lines before the schedule crisis.

Discussion: unreachable or just ungenerated?

Name the evidence that would settle the decision.

- A legal bin has never been hit.
- The sequence library does not generate it.
- Is a structural exclusion justified?

A dashboard is a set of instruments

No single curve determines readiness.

- Read per-feature status with coverage trends.
- Correlate bug curves, regression health and debug effort.

A flat bug curve is ambiguous

Clean behavior and stimulus starvation can look alike.

- Check freshness and breadth of stimulus.
- Inspect checker integrity and where recent bugs were found.

Sign-off is an evidence decision

Conditional release must name the accepted gap.

- Discharge plan rows and review exclusions.
- Record reproducible baselines and bug dispositions.
- A conditional decision needs an owner and expiry.

Discussion: a late severe bug

Name the evidence that would settle the decision.

- Discovery had been flat.
- A new checker finds a severe unmodeled failure.
- Which earlier conclusion must be revisited?

Exercise: coverage without comparisons

Decide what the report can establish.

- The modeled space is fully hit.
- No output comparison was performed.
- State the missing evidence and a defensible closure decision.

What to change on Monday

Make one existing verification decision reviewable.

- Review one row with mixed closure criteria.
- Trace a coverage hit to its checker.
- Challenge one exclusion or dashboard inference.

Scope and next reading

Use the handout for the detailed artifacts and sources.

- Read the Planning and measurement handout and book Chapters 5–7.
- Tool configuration and full coverage syntax are outside this lesson.